

Worksheet 12 – Understanding Baker's gamma correlation

The distances between pairs of 5 objects are given:

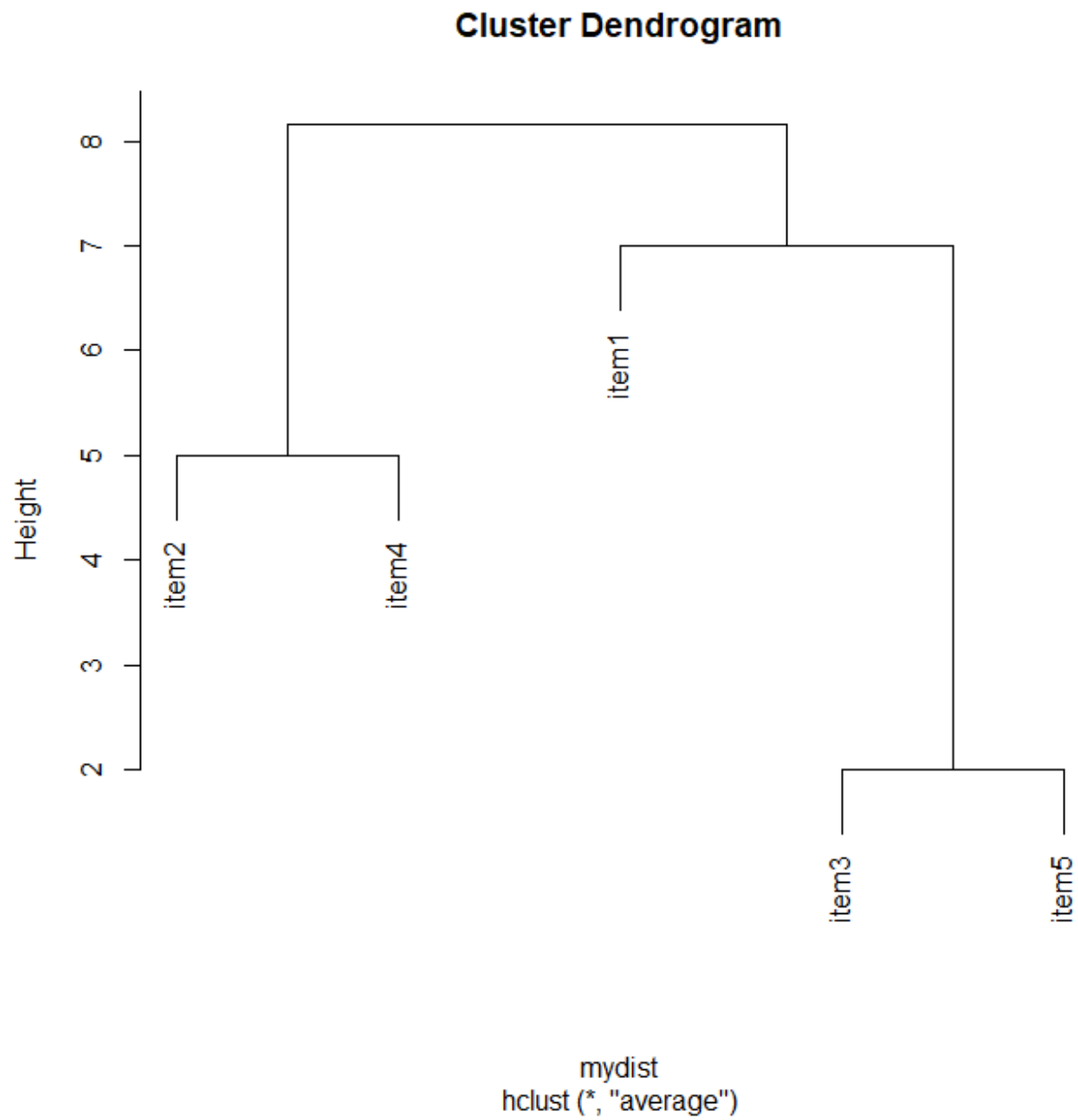
	1	2	3	4	5
1	0				
2	9	0			
3	3	7	0		
4	6	5	9	0	
5	11	10	2	8	0

Study the code below and answer the questions that follow:

```
> item1<-c(0,9,3,6,11)
> item2<-c(9,0,7,5,10)
> item3<-c(3,7,0,9,2)
> item4<-c(6,5,9,0,8)
> item5<-c(11,10,2,8,0)
> mydist<-as.dist(rbind(item1, item2, item3, item4, item5))
> as.matrix(mydist)
      item1 item2 item3 item4 item5
item1     0     9     3     6    11
item2     9     0     7     5    10
item3     3     7     0     9     2
item4     6     5     9     0     8
item5    11    10     2     8     0

> hc1 <- hclust(mydist, method = "average")
> hc2 <- hclust(mydist, method = "ward.D2")
> dend1 <- as.dendrogram (hc1)
> dend2 <- as.dendrogram (hc2)
> dend_list <- dendlist(dend1, dend2)
> # Cophenetic correlation coefficient
> cor_cophenetic(dend1, dend2)
[1] 0.8651635
> # Baker correlation coefficient
> cor_bakers_gamma(dend1, dend2)
[1] 0.503876
>
```

```
> hc1$height  
[1] 2.000000 5.000000 7.000000 8.166667  
  
> hc1$order  
[1] 2 4 1 3 5  
  
> plot(hc1)
```



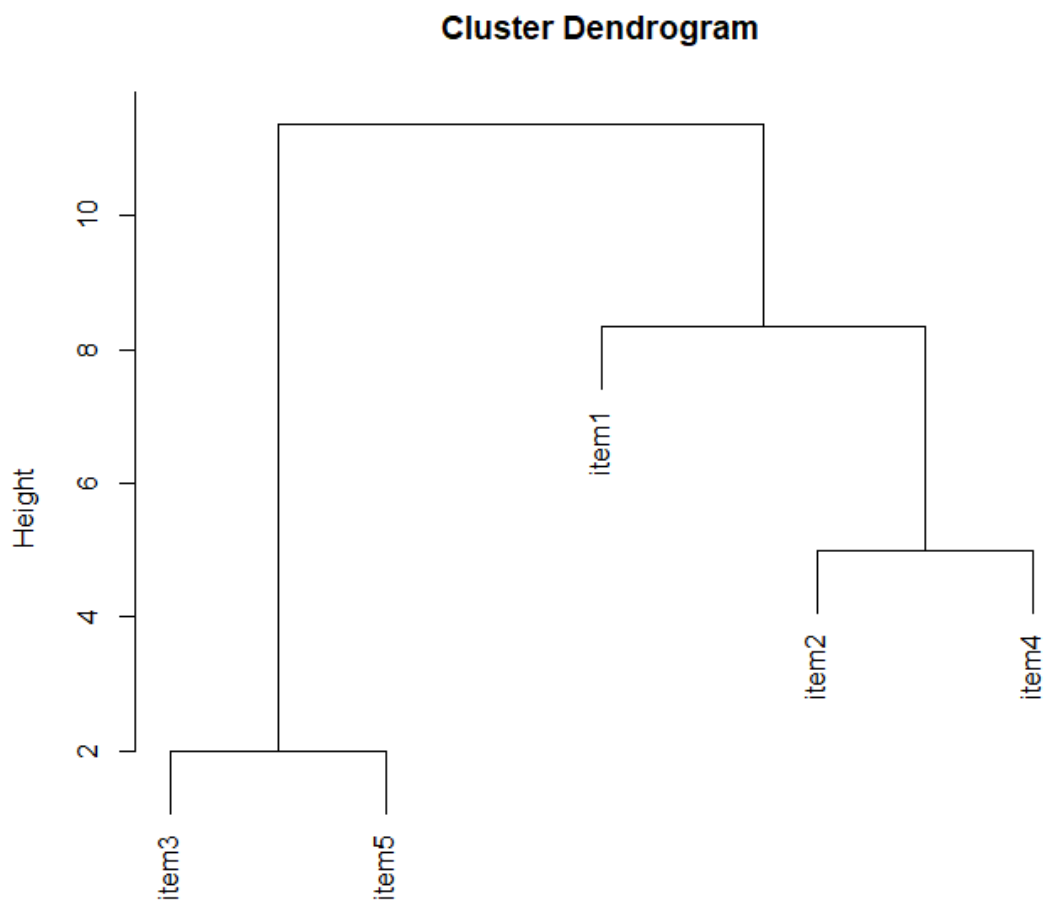
```
> hc2$height
```

```
[1] 2.000000 5.000000 8.346656 11.372481
```

```
> hc2$order
```

```
[1] 3 5 1 2 4
```

```
> plot(hc2)
```



mydist
hclust (*, "ward.D2")

(a) Show how `cor_cophenetic(dend1, dend2)` obtained a value of **0.8651635**.

(b) Show how `cor_bakers_gamma(dend1, dend2)` obtained a value of **0.503876**.

Some useful code and results to check answers:

```
> x<-c(8.166667, 7, 8.166667, 7, 8.166667, 5, 8.166667, 8.166667, 2, 8.166667)
> y<-c(8.346656, 11.372481, 8.346656, 11.372481, 11.372481, 5, 11.372481, 11.372481, 2, 11.372481)
> cor(x,y)
[1] 0.8651635
> x<-c(1,2,1,2,1,3,1,1,4,1)
> y<-c(2,1,2,1,1,3,1,1,4,1)
> cor(x,y)
[1] 0.8019802
> cor.test(x,y,method="spearman")
```

Spearman's rank correlation rho

data: x and y

S = 81.86, p-value = 0.1376

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.503876

Warning message:

In cor.test.default(x, y, method = "spearman") :

Cannot compute exact p-value with ties